

Finite Agent Machines

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Abstract

TBD

1 Introduction

Giving an AI an open-ended goal to run with creates a fundamental problem of trust: the same system decides what is worth pursuing and what will count as success. In exploratory work, we do not yet know what we will find or which findings will matter. An investigation of an unfamiliar software system, for example, may begin with the aim of identifying worthwhile improvements, while leaving the relevant problems and possible solutions to be discovered. Evidence gathered during the investigation helps determine what to pursue and what would count as a useful result. A concrete end state and a complete criterion for success may therefore be unavailable before the work begins. Even when the desired outcome is known, verifying that the outcome has been achieved may be costly or inconclusive.

Exploratory work can be understood as a search whose concrete objectives emerge and change as evidence accumulates. We introduce *nondeterministic finite agent machines* (FAMs)¹, a model for process-driven agentic teams that makes the search process explicit and interpretable by defining how agents investigate possibilities, assess findings, and decide what to pursue next. FAMs ensure that work proceeds in a permitted order and make the path to a result inspectable. Confidence in the result can then draw on the reliability of the search process and evidence that it was followed, even when the result itself could not be specified in advance or conclusively verified.

The same FAM can therefore produce different executions. An agentic transition may divide a problem into several investigations or produce findings that enable different continuations, while the available types of investigation and continuation remain defined by the FAM. A FAM defines a finite repertoire of predefined agentic transitions, each admitting multiple possible outcomes. Agents determine how those transitions are applied to the work at hand, allowing the search to develop through discoveries without requiring its concrete branches to be enumerated in advance.

Here, *finite* refers to the set of authored transition schemas. Finite transition schemas do not imply finitely many artifact states: executions can accumulate new artifact files that affect subsequent work. The number of concrete branches and iterations need not be fixed in advance.

Recent projects such as Compound Engineering [3], Compound Writing [4], and Research–Plan–Implement (RPI) [1] organize agent work through reusable skills and staged procedures. These approaches make a process explicit, but procedural instructions expressed in prompts do not by themselves guarantee adherence. Where a coordinating agent selects activities, judges whether their requirements have been met, and decides what happens next, the process remains dependent on that agent’s interpretation. An agent may omit a prerequisite, advance on insufficient evidence, or allow an unsupported assumption to shape subsequent work. FAMs address this coordination problem by making declared dependencies conditions for execution, so that following the prescribed process does not rely solely on the coordinating agent’s willingness or ability to follow instructions.

We implement this model in Alinery², where a Playbook specifies a particular FAM and the Playbook engine executes it. A single Playbook can produce many different executions, each developing according to the evidence generated during the work. The remainder of this paper develops the FAM model and its execution semantics, relates it to existing approaches, and describes its implementation in Alinery.

2 Finite Agent Machines

A Finite Agent Machine (FAM) defines an abstract state space, a finite set of agentic operations, and rules governing when those operations may run. An execution engine enforces these rules, providing hard guardrails on how the agents’ work is coordinated. A state is a set of available artifact filenames; agents interpret and create the files’ contents. An execution engine binds operations to available inputs, records the resulting executions, and enforces the conditions for passing their outputs to other agents. A single definition can therefore generate different numbers of executions and different chains of evidence without changing its authored operations.

¹Intentionally echoes finite state machines

²<https://github.com/deleganceai/alinery>

Definition 1 (Finite Agent Machine). A FAM is a transition system

$$\mathcal{F} = (\mathcal{A}, \mathcal{S}, s_0, \rightarrow),$$

where $\mathcal{A}$ is a finite set of authored agent operation schemas, $\mathcal{S}$ is a set of artifact states, $s_0 \in \mathcal{S}$ is the initial state, and $\rightarrow \subseteq \mathcal{S} \times \mathcal{A} \times \mathcal{S}$ is the transition relation. Each state is a finite set of available artifact filenames. A transition $s \xrightarrow{\alpha} s'$ represents a completed invocation of operation α using artifacts named in s and publishing output artifacts whose filenames form a set P , so that $s' = s \cup P$.

Here, *finite* refers to the authored operation set $\mathcal{A}$. The same operation may publish different sets of artifact filenames from the same state, giving rise to more than one possible successor. Repeated invocations can add new filenames, so the number of reachable states may be unbounded and an execution need not terminate.

Artifacts and States

An *artifact* is a file that carries information used or produced by an agent operation, such as a task description, source assessment, or proposed solution. Let $\mathcal{D}$ denote the universe of artifact filenames. Each state $s \in \mathcal{S}$ is a finite subset of $\mathcal{D}$. The initial state s_0 contains the filenames of the initial input artifacts.

Artifacts with different filenames remain distinct even when they serve the same purpose or contain identical text. Reading an artifact does not remove its filename from the state, so the same artifact can support several later operations.

Agent Operations

An *agent operation schema* $\alpha \in \mathcal{A}$ is a reusable specification of work for an agent. We write

$$\alpha = (I_\alpha, O_\alpha, \xi_\alpha).$$

The input and output specifications are families of permitted filename sets,

$$I_\alpha, O_\alpha \subseteq \mathcal{P}_{\text{fin}}(\mathcal{D}),$$

where $\mathcal{P}_{\text{fin}}(\mathcal{D})$ denotes the finite subsets of $\mathcal{D}$. The execution specification ξ_α supplies the agent's instructions and execution configuration.

An *invocation* is one execution of a schema. Its input filenames must form a set in I_α and all be present in the current state. The agent interprets the corresponding files under the specified instructions, and its published output filenames must form a set in O_α . Neither specification predetermines file contents. The same schema, α , can be invoked repeatedly with different inputs, so additional executions do not enlarge the authored operation set $\mathcal{A}$.

Agentic Transitions

For a completed invocation of schema α from state s , let B be its set of input filenames and P its set of published output filenames. The invocation must satisfy

$$B \in I_\alpha, \quad B \subseteq s, \quad P \in O_\alpha, \quad P \neq \emptyset,$$

and produces the *agentic transition*

$$s \xrightarrow{\alpha} s', \quad s' = s \cup P.$$

All filenames in s remain available in s' .

Each invocation publishes one or more artifacts, whose filenames form P . By choosing which permitted outputs to publish, the agent can satisfy the input requirements of different subsequent operations, enabling conditional branching. Published artifacts can support several continuations, while fresh artifacts that supply inputs to an earlier operation can enable another iteration.

Executions

A *rollout*, or execution of a FAM, is a finite or infinite sequence of transitions beginning at the initial state s_0 :

$$s_0 \xrightarrow{\alpha_1} s_1 \xrightarrow{\alpha_2} s_2 \xrightarrow{\alpha_3} \dots,$$

where each $\alpha_i \in \mathcal{A}$ and each transition belongs to the relation $\rightarrow$.

The same schema may appear repeatedly in a rollout, and different rollouts of the same FAM may follow different sequences of transitions. Repeating an operation can extend the accumulated state with new artifact filenames.

3 Motivating Examples

To illustrate how a FAM organizes agent work, consider a team conducting a literature review. Given a research question, the agents find relevant papers, assess their findings, and synthesize a report. Figure 1 shows three FAM definitions: one that branches across papers, one that loops as new sources are discovered, and one that combines both structures.

Branching

The *Find sources* operation selects relevant sources and publishes a source-request artifact for each. Each request enables a separate invocation of *Read source*, which publishes an assessment. *Synthesize review* combines the assessments into a review. The sources selected during execution determine which reading branches occur; the same reading schema serves every selected source.

Looping

The second version examines one source at a time. After *Find source* and *Read source*, *Synthesize review* incorporates the new assessment into the accumulated review. *Assess next step* chooses between two permitted outcomes: a fresh request to investigate a promising citation or unresolved question, or a closing report. A fresh request enables another pass through the same operations, with earlier findings carried forward. The next source and the number of passes are not fixed in advance.

Branching and Looping

The third version repeats the branching process. A round can assess several selected sources, and synthesis combines their assessments with earlier findings. *Assess next step* may publish a fresh request for the questions or citations that warrant

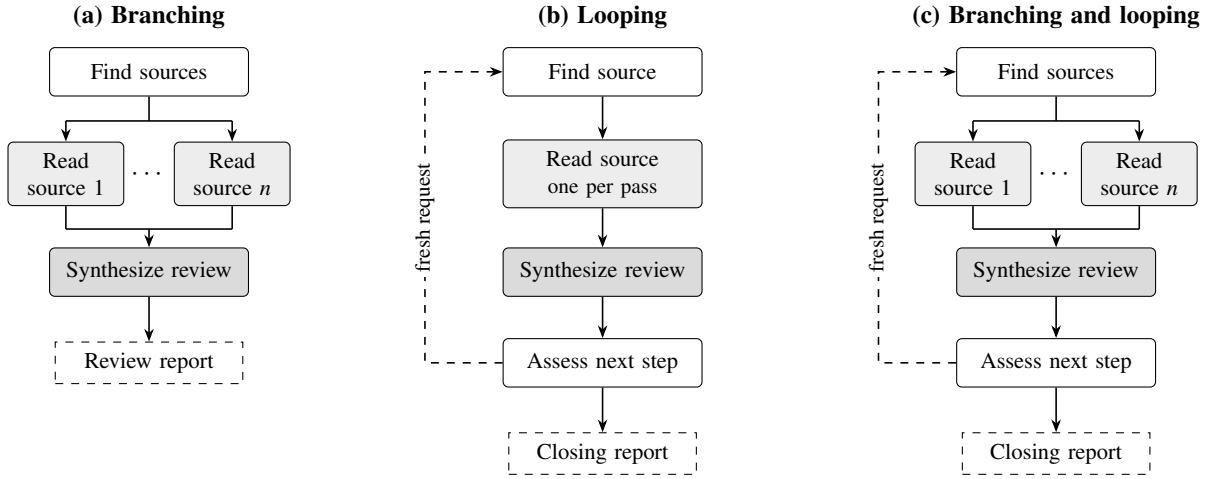


Figure 1: Session patterns specified by three literature-review FAMs. Each rounded box represents an agent session invoking the named operation. In (a) and (c), there is one reading session per selected source; the ellipsis leaves the source count unspecified. These sessions invoke the same *Read source* schema. Solid arrows show principal artifact dependencies. Dashed return arrows indicate new passes enabled by fresh requests; dashed rectangles are output artifacts.

more investigation, enabling another round, or a closing report. Figure 2 shows one possible rollout with three sources in the first round, two in the second, and one in the third before the review concludes.

4 Background

Why Software Verification Is Hard

the main question is whether the benefit of an improvement to a known solution to a problem is worth the cost of searching the space of possible solutions (cite alpha evolve here, and open evolve). For many problems in research and software development, complete verification is unattainable, and this limits the ability to apply LLM-based agents, because every potential solution generated by the agents produces more verification work, and this becomes prohibitively expensive.

There’s other issues here that are relevant... while we want agents to be all powerful, the reality is that they still have a number of limitations in practice. Context window sizes, following instructions generally, especially on problems in domains for which they have not been trained well. (this is why benchmaxxing is a thing). The scope of a problem given to an agent is incredibly important. From a practitioners point of view, small scoped tasks are much more likely to be executed well by an agent than scope involving larger tasks. This is why the notion of ‘sub-agents’ have become so popular. Sub-agents are a way to pass out narrow focused tasks so that each agent has a small piece of work to focus on.

Most sub-agents in use today refers to an automatic feature driven by a larger model.

When verification is prohibitively expensive, we need to reduce the number of candidate solutions generated by agents, so that verification costs are within bounds of the budget of an individual or team. On this spectrum, the most

extreme version is for all code to be manually reviewed by a human, and on the other end of the spectrum are techniques to limit the space of what agents can do.

We present a novel multi-agent architecture to solve alignment and verification problems with Large Language Models and similar neural-based models.

The ideal use case is work that cannot be trivially or automatically (including with software) verified. Additionally, the consequence of getting the work right is high stakes. For that reason, we avoid the problem of prototyping. We also are interested in work that is not possible to verify the outcome after the fact. Therefore, since the outcome of applying AI to solve a problem is not completely verifiable, the process that produced it offloads some of the trust that a human puts into verifying it. If they can verify the process that was followed by the AI, then they can reduce the amount of verification needed on the end result.

There is a spectrum of approaches to perform concrete work. You could create a single agent, you could use dynamic workflows like Claude Code, you could do everything by hand yourself.

Petri Nets and Concurrent Agent Execution

Petri nets describe processes in which several activities may be enabled at once and later activities depend on combinations of earlier results. A net contains *places*, which hold tokens, and *transitions*, which consume tokens from input places and produce tokens in output places. The distribution of tokens across all places is the *marking*, or global state of the net. A transition is enabled when its required input tokens are available; independent transitions can proceed without imposing an order between them, while a transition requiring several inputs provides a synchronization point [8].

Colored Petri nets associate typed data values with tokens, allowing a transition to distinguish particular work items and

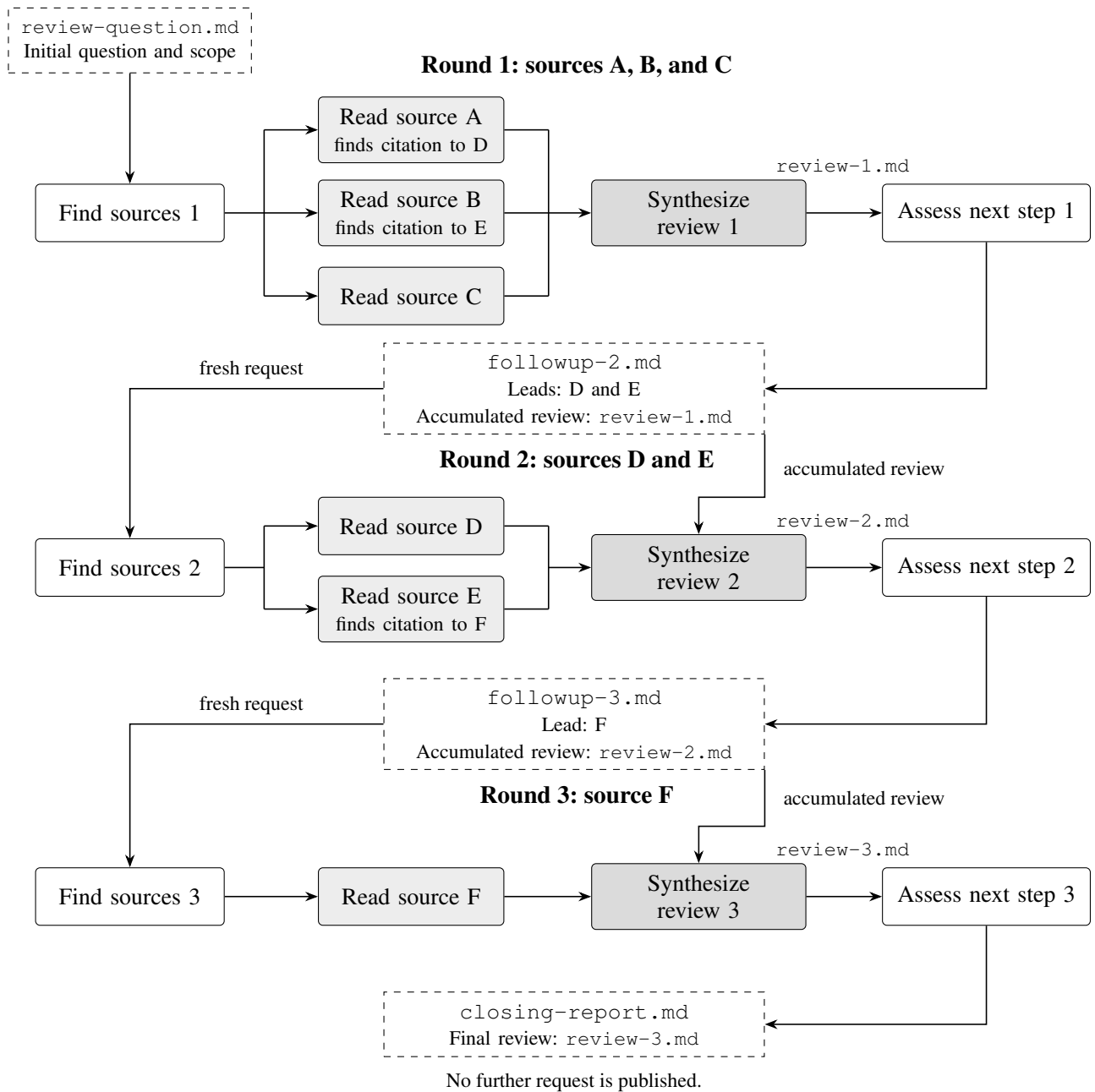


Figure 2: One possible rollout of the branching and looping FAM in Figure 1(c). Four authored operation schemas produce 15 completed invocations across three rounds. Each rounded box is one invocation; dashed boxes identify artifacts. Reading connections represent the request-X.md and assessment-X.md handoffs for source X. Fresh requests enable further rounds until a closing report is published. Arrows show principal artifact dependencies. Intermediate states are omitted; accumulating output filenames in completion order gives the state-transition view of Section 2.

match their associated inputs [8]. Read arcs add a complementary operation: requiring a token to be present without consuming it [11]. Reusable information can therefore enable several activities, while ordinary consuming arcs can represent resources whose use must be exclusive.

Playbooks implement an artifact-driven concurrent execution model closely related to colored Petri nets with read arcs. A Playbook defines agentic operations called Steps and connects them through declared artifact inputs and possible outputs. Each execution binds a Step to particular artifact occurrences, retaining the identity of each publication and its producing execution. In a Petri-net interpretation, occurrences provide data-bearing tokens, input requirements determine enabling conditions, and accepted publications supply new tokens. For example, two reviewers may read the same brief independently, while a synthesis Step requires both review results. Read arcs express reuse of the brief, and the synthesis inputs express the AND dependency. Occurrence and provenance information distinguishes results from different requests or loop passes.

The engine also records behavior that a faithful net model would need to make explicit. An execution occupies capacity while active, and its accepted outputs become available to dependent executions only after shutdown is confirmed. Recorded input bindings prevent duplicate automatic executions; session limits and exclusive coding access constrain launches. Collection inputs support one execution per member and synchronization over a nonempty accepted collection after all expected contributors and source obligations have completed. Agent outcomes introduce nondeterminism in artifact contents and in the permitted outputs published. Publication of the required artifact roles determines downstream eligibility; agents interpret artifact contents.

The Petri-net comparison identifies a structural relationship rather than an established formal equivalence. An exact encoding would need to preserve occurrence matching, binding deduplication, collection closure, and execution lifecycle rules. The distinction between a finite net description and its reachable markings also clarifies finiteness: a finite set of Step definitions can generate an unbounded number of executions and artifact occurrences. A limit on simultaneous sessions does not by itself make the complete execution state space finite.

Related Work and Positioning

Classical planning supplies precise vocabulary for states, action applicability, and successor transitions [5]. Hierarchical task network planning supplies authored task decompositions that a planner refines toward executable networks [2]. Reactive Playbooks use neither goal search nor search over alternative Step schemas or task networks: the process definition is fixed, while agent-produced data determines how many fixed Step executions are late-grounded. The model is therefore closer to execution and monitoring of an authored process than to plan synthesis.

Workflow languages commonly represent splits, joins, choices, and loops as control-flow patterns [12]; statecharts likewise make control state and transitions explicit [7]. A Playbook may be rendered with similar shapes, but its source

of truth is exact material dependency and provenance. The definition view may contain a visual cycle, whereas the execution view contains only newly published DAG nodes. Avoiding a cursor is not a claim that workflow formalisms are incapable of the example. Material-state sufficiency is a design choice that makes it possible to reconstruct why each agent execution occurred.

Temporal planning for actions with duration distinguishes an action's interval of execution from the points at which its effects change the state [6]. A Playbook Step execution draws a narrower analogous boundary: several sequential harness Sessions may perform work, but accepted completion exposes one atomic successor. The paper does not adopt the temporal constraints, time-varying numeric state variables, or formal plan validation of the Planning Domain Definition Language (PDDL).

Multi-agent frameworks such as AutoGen provide flexible conversation and tool interaction patterns [13]. A conversational agent can run inside a Playbook Session; the Playbook model instead governs durable publication between executions. General provenance models describe entities, activities, agents, and derivations [10]. Playbook provenance specializes that vocabulary with exact input bindings, fulfillment relationships, automatic claim identity, and coverage-checked state parents.

Previous Literature Review Example (Preserved)

A literature review often reveals sources that we did not know to search for. Reading one paper may uncover a relevant citation, and reading the cited paper may uncover another. We can define how to search, evaluate sources, and incorporate the findings without knowing in advance which papers the review will include or how many rounds of discovery it will require. Alinery's Academic Survey Playbook expresses this process through eight operations, with parallel work within each round and a loop for following new citations. Operations are connected through artifact dependencies: one operation can publish an artifact that another requires as input. An execution can begin only when all its required inputs are available. We call one pass from planning through continuation a wave.

Suppose we want to review how agents retain and reuse information over long tasks. The initial task description is supplied as `ticket.md`. A framing agent first writes `review-brief.md`, establishing the question, scope, and criteria for including sources. Later waves continue to use this original brief. A planning agent reads the brief and the current `ticket.md`, then writes a search plan and three search requests: one on retrieving stored notes, one on summarizing past interactions, and one on maintaining task state. Once the planner completes its handoff, the engine can run a separate gathering agent for each request, subject to the task's concurrency limit. Three requests produce three executions of the same gathering operation; the Playbook does not need three authored copies.

Each gathering agent publishes a search report, including any access limitations or an absence of candidates. The source-selection operation waits for all three reports before comparing the candidates and removing duplicates. Suppose the selection agent identifies seven distinct sources to read and publishes seven source requests. The engine then creates seven reading executions, each bound to one request and the shared review brief. Each reader evaluates its assigned source and publishes an assessment containing findings, limitations, and potentially useful citations. Search requests and source requests therefore provide two fan-outs whose sizes are determined by agents during execution.

The evidence-consolidation operation merges the source assessments. Six completed readers are insufficient while the seventh remains outstanding: the engine waits for the complete set of assigned readers to finish their handoffs. A reader that cannot access its source can report that limitation in an assessment; an unfinished reader cannot be silently dropped from the merge. The consolidation agent compares the assessments, updates the cumulative evidence record, and identifies citation leads that have not already been considered. A drafting agent uses that evidence to produce a survey draft. The engine establishes which assessments belong to the wave and when the merge can run; the agents judge source relevance, interpret findings, and decide which claims the evidence supports.

Suppose two readers each identify one promising citation outside the seven sources already read. The continua-

tion agent writes a new `ticket.md` naming those citations and carrying forward the evidence record, prior assessment references, and current draft. After that handoff, the engine enables another planning execution using the new ticket and the original review brief. The framing operation does not repeat. In this illustrative second wave, the planner writes two search requests, one for each new citation, and gathering agents locate the two cited sources. The source-selection agent checks both against the accumulated record and publishes two new reading requests. The resulting assessments support a revised survey.

Suppose source H, read in the second wave, cites another relevant source J. The continuation agent publishes a new ticket for J, and a third wave plans one search, locates J, and assigns one reader. Consolidation incorporates J's assessment into the cumulative evidence, and the drafting agent updates the survey again. The third wave adds new executions of the same operations and retains the evidence accumulated in the earlier waves.

The continuation agent also provides a way to conclude the review. When no useful leads remain, an agreed search boundary has been reached, or further work needs direction, the agent presents the current draft and evidence coverage to the human. In this rollout, the human concludes the review after the third wave. The agent then publishes `review-stopping-report.md` without another ticket. That publication enables no further search wave. The closing decision records the scope and limitations of the review; it does not establish that every relevant source has been found.

The eight operation definitions remain fixed throughout this execution, while discoveries determine the number of searches, source readings, and successive waves. Every wave has explicit merge conditions, and each new ticket retains a connection to the evidence that motivated further work. Agent instructions require relevance judgments and avoidance of duplicate readings; the engine enforces the declared artifact dependencies. The FAM therefore permits an expanding investigation while retaining an inspectable process for admitting work and incorporating its results.

Previous Draft (Preserved)

Abstract

Long-running agent work produces both a result and an execution process: choices about decomposition, evidence, parallelism, combination, and when no further work is enabled. Synthesizing that process anew makes its decomposition and coordination choices another task-specific output. For recurring work whose results are difficult to check directly, a versioned process definition and an exact execution record give reviewers stable objects to inspect. The technical challenge is to retain reusable structure without forcing every execution down a fixed path.

We present *Reactive Playbooks*, a material-state execution model that separates revision-stable agent task templates, called Steps, from their concrete applications determined at runtime. Exact artifact occurrences and immutable collection members determine which Steps become eligible. Each execution works in isolation and may publish one validated successor state; parallel executions publish siblings, coverage-checked merges record all supplying parents, and follow-up publications reuse Step definitions while extending a forward-only history.

A research-to-design Playbook demonstrates runtime-determined parallel appraisal, question-level synthesis, explicit design merge, and recurrence. An event-driven reference algorithm yields obligations for atomic visibility, completion-order-independent lineage, at-most-once automatic claims, coverage-preserving merge, and forward recurrence; it also defines quiescence as the absence of eligible, queued, or running work. Together, the model and obligations let one versioned definition generate materially different executions while preserving an exact account of which activities ran, the material each activity consumed, and how each result became visible.

1 Introduction

An AI system asked to investigate a production failure, conduct research, or design software must do more than produce a final artifact. The system must instantiate an execution process: decompose the work, decide which evidence to seek, determine what can run in parallel, combine intermediate results, and decide when no further work is enabled. When the final artifact is difficult to evaluate directly, a reviewer also needs to understand that execution process. An execution record cannot prove the artifact correct, but it can expose which activities ran, what material each activity consumed, and how each result became visible.

The execution process may be synthesized for one task or deliberately retained as a versioned definition for repeated use. Task-specific generation avoids the cost of maintaining a reusable procedure when work is novel or one-off. This paper studies the complementary case in which recurring process decisions remain explicit across executions. The distinction concerns lifecycle rather than authorship: either a person or an agent may draft the process. A reusable definition also need not impose a fixed trace; the retained Step definitions can remain stable within a revision while runtime-produced material determines which concrete activities run, fan out, merge, recur, or leave no work enabled.

Retaining the process definition exposes an execution-semantics problem. Agent frameworks can compose models, tools, and humans into useful interaction patterns [13], while long-running repository work may require one activity to edit for minutes, exhaust a model context, continue in another model-harness invocation (a *Session*), or finish concurrently with another activity. A saved diagram or set of prompts alone does not determine when intermediate edits become shared, how concurrent results relate, which evidence authorizes a merge, why an earlier Step becomes eligible again, or how reevaluation avoids creating the same application twice. Ambiguous publication, lineage, recurrence, and claim semantics become acute when each activity is a fallible, long-lived agent operating on material files rather than a deterministic function over a small symbolic state.

To make the execution problem concrete, we use a research-to-design Playbook whose purpose is to turn an initial research agenda into an evidence-linked design recommendation. Frame derives decision-relevant questions; Discover identifies candidate sources; Appraise evaluates each question-source pair; Synthesize combines the appraisals for one question; and Design integrates the question-level findings. Discovery may create any finite number of appraisal requests, and Design may publish a targeted follow-up question when another research pass could change the design. The example therefore combines a runtime-determined number of branches, parallel work, explicit merges, and recurrence.

Reactive Playbooks address this execution-semantics problem with a material-state semantics. A Playbook revision contains a validated set of Step definitions. *Grounding* instantiates one Step definition only after exact artifact occurrences—particular publications, even when their bytes match earlier versions—or immutable collection members resolve its input declaration. Before dispatch, a durable claim reserves the definition revision, exact binding, and runtime-policy scope so that the same automatic application is not created twice. The resulting Step execution may span sequential Sessions, but its edits remain private until an explicit completion request passes validation and atomically publishes one immutable state. Every publication wakes evaluation over the recorded history; no active cursor, routing token, or loop counter determines what runs next. One Playbook revision can therefore produce materially different executions while retaining exact material lineage for each execution.

We make three contributions. First, we give a compact execution model that separates readiness evidence, bound inputs, candidate states, selected parents, and recorded lineage. Second, we use one evolving research-to-design Playbook to show how material dependencies produce linear order, late-grounded parallelism, explicit merging, and recurrence while concrete history remains a directed acyclic graph (DAG). Third, we define an event-driven grounding, claiming, and publication algorithm and derive implementation-neutral obligations for a conforming runtime. Together, the model, example, and reference algorithm provide an implementation-neutral foundation for executing versioned definitions of agent processes across paral-

lelism, merging, and recurrence while preserving exact material provenance.

2 Planning Policies and Persistent Execution

In nondeterministic planning, an action may have more than one possible result, and a policy associates encountered states with actions so execution can respond to the result that actually occurs [9]. Reactive Playbooks are policy-like only in this persistent, reactive sense. A Playbook revision fixes Step schemas—their instructions, runtime configuration, input and readiness declarations, output contracts, and collection relationships—while execution grounds concrete applications from immutable material history. Unlike a conventional planning policy, evaluation may enable several Step applications concurrently, consults provenance and durable claims rather than one active state, and has no goal test or optimization objective. This planning-policy analogy is distinct from the runtime policies for capacity, retries, authority, and storage discussed later.

The execution horizon—the number of concrete Step executions that unfold—is not fixed. Continued publication of fresh bindings can produce arbitrarily long and potentially nonterminating forward histories even though a Playbook revision contains finitely many Step definitions. Planning distinguishes an intended goal state from a *dead end*, a non-goal state from which the goal is unreachable. The Playbook model defines neither category because it has no single active state and no goal predicate. Its closest stopping condition is history-wide quiescence: no binding is eligible and no claimed work is queued or running. Quiescence may coincide with an application’s intended terminal result or with an unproductive stopping point, but the runtime does not decide which; it remains attached and may wake after a later publication or manual event.

3 One Playbook, Increasing Structure

Throughout the paper, the running Playbook begins with a research agenda and produces an evidence-linked design recommendation. Frame turns the agenda or a follow-up handoff into questions; Discover selects plausible question-source pairs at runtime; Appraise evaluates those pairs independently; Synthesize combines the declared appraisals for each question; and Design merges the question-level conclusions. If Design identifies a material unresolved question, it publishes a follow-up handoff and the same five Step definitions apply again; otherwise the Playbook becomes quiescent.

Figure 3 gives a structural overview of this Playbook at three resolutions; Listing 1 in the appendix records its compact schematic definition. Panel (a) introduces the linear material dependencies; panel (b) exposes runtime-grounded Appraise siblings and their explicit Synthesize merge; and panel (c) adds the recurrence arc created when Design publishes a follow-up that grounds another Frame execution. Solid arrows denote material dependencies rather than authored routing. The dashed arc denotes reuse of the same Step definitions over new bindings, not a backward edge in recorded state history.

A linear first reading

At its coarsest resolution, the running Playbook forms a material-dependency chain: Frame produces questions, Discover produces a source map, Appraise produces evidence, Synthesize interprets the evidence, and Design applies the interpretation. Ordering is a consequence of material availability. Synthesize does not follow Appraise because an edge moved a token; Synthesize becomes groundable because exact appraisal-result occurrences now fulfill its declared input set. The next subsection refines Appraise and Synthesize into collection-grounded fan-out and merge.

The linear reading gives two basic requirements. Intermediate writes must not make downstream Steps applicable, because an agent can revise or abandon them. Conversely, accepted work must have an exact identity, because a filename such as `design.md` may be republished with identical or different bytes during a later pass. The model therefore distinguishes an artifact *occurrence*, identified by its creating execution and state, from its content digest.

Dynamic branching and explicit merging

Discover cannot know the useful fan-out before it inspects the questions and search space. At accepted completion, Discover freezes a finite `appraisal-requests` collection. Each member binds one Appraise execution, so four members yield four independent executions without four authored copies of the Step. The scheduler may run these executions concurrently, subject to a capacity policy that does not change their logical bindings.

A question-level Synthesize execution requires the successful results that fulfill its exact subset of appraisal requests. The present model also requires completion of the entire Discover expansion as a conservative readiness barrier, without treating unrelated appraisal states as inputs or parents. At completion, Synthesize selects a duplicate-free set of authorized states whose result occurrences collectively cover the bound requests. The engine, not the agent, verifies coverage and records those states as parents.

Repetition without cyclic history

Design emits a finite `followup-research` collection. A nonempty collection creates a new exact binding for Frame and therefore another forward research pass. An explicitly empty collection creates no Frame binding. Once no other binding is eligible or running, the Playbook is *quiescent*. Quiescence means that no configured work is currently available; it is not a proof that the design is correct, a goal has been achieved, or later human work cannot wake the Playbook. If every Design execution instead publishes a fresh follow-up member, forward recurrence can continue without a fixed bound even though the recorded history remains acyclic.

4 Execution Model

Immutable task states

Let $\mathcal{N}$ be exact artifact names, $\mathcal{O}$ immutable artifact occurrences, and $\mathcal{W}$ material workspace checkpoints. A sealed

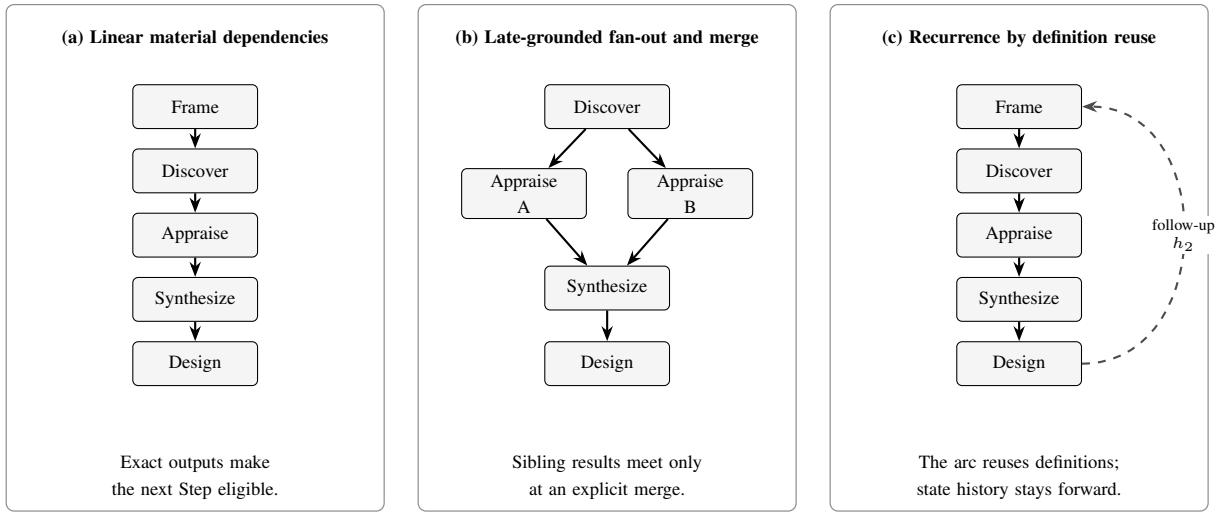


Figure 3: Three structural views of the running research-to-design Playbook. Panel (a) shows the linear dependency chain created by exact outputs. Panel (b) shows Discover late-grounding independent Appraise executions and Synthesize merging their complete result set. Panel (c) adds a dashed loop arc from Design to Frame: follow-up h_2 makes the same Frame definition applicable to a new binding. The arc depicts recurrence in the Playbook definition; every accepted execution still adds a new state along forward-only recorded history.

task state is

$$s = (id_s, w_s, A_s, C_s, Q_s, \rho_s), \quad (1)$$

where id_s is a fresh state identifier and $w_s \in \mathcal{W}$ is a complete checkpoint. $A_s : \mathcal{N} \rightarrow \mathcal{O}$ is a partial map from artifact names to visible occurrences, so a name may be absent; C_s maps collection names to finite frozen member sets; Q_s is a duplicate-free set of already sealed parent-state identifiers; and ρ_s records the producing execution, definition revisions, and bound-input provenance. A state is immutable after publication. Several states may coexist; lineage does not mark one state as the active truth.

An occurrence identifies a publication event, not only bytes. Republishing unchanged bytes can create a new occurrence, while an inherited artifact keeps its prior occurrence identity. Occurrence identity lets a binding state exactly which work it consumed without making every descendant appear fresh.

The state history $H = (S, E)$ contains edge (p, s) for every $p \in Q_s$. Because publication may name only already sealed parents and a new state receives a fresh identity, H is acyclic by construction. An ordinary execution has one mechanically fixed parent. A merge-capable execution may select one authorized parent when that state already covers every bound input; selecting several distinct authorized parents creates a merge transition when the inputs live on siblings.

Definitions, bindings, and collections

A Playbook revision P^r is a finite set of validated Step-definition revisions. A definition d^r contains agent instructions, runtime configuration, input declarations, readiness conditions, an output contract, and collection relationships.

Validation rejects unresolved references and incompatible collection keys before any Session starts.

A binding b contains exact ordinary occurrences and collection-member identities. Three grounding forms are sufficient for the running example:

1. an ordinary input resolves an exact named occurrence in one sealed state;
2. `each(C)` enumerates one candidate grounding for every exact member of frozen collection C ; and
3. `complete-set(R)` resolves the exact successful results whose immutable fulfillment records cover required member set R .

Collection membership is never inferred from filename patterns, directory scans, or natural-language parsing. The producer explicitly declares a finite collection during completion; the engine validates and freezes the declaration inside the successor state.

Accepted publication records each declared fulfillment as an exact relation from a result occurrence to the request member it fulfills. Failed or rejected executions record no successful fulfillment. Claims, rather than grounding, suppress duplicate automatic executions for already claimed exact bindings.

Readiness and input resolution are separate operations. Let u be a readiness witness set and b the exact input binding. A broad condition may establish that all Appraise requests from one Discover expansion have some successful result, while a Synthesize binding contains only the results for one question. The witness states in u do not become input, merge candidates, or lineage unless their occurrences also appear in b .

Automatic duplicate suppression uses a durable claim

identity

$$\kappa = \text{ClaimIdentity}(P^r, d^r, b, \chi), \quad (2)$$

where χ is the still-open policy scope, such as ordinary-parent context or an ancestry scope. Its canonical storage encoding is an implementation choice. Claim acquisition and queued-execution creation form one atomic operation. A claim names logical identity; it is not a proof of successful completion. Explicit manual reruns receive distinct execution identities. Failure retention and automatic retry policy remain separate decisions rather than hidden parts of the binding.

Executions and Sessions

A Step execution

$$e = (P^r, d^r, b, A_e, K_e, o_e, W_e, \langle q_1, \dots, q_k \rangle) \quad (3)$$

binds one definition revision to one exact input, a frozen authorized candidate set A_e , required coverage K_e , a mechanically fixed ordinary parent o_e when applicable, an isolated workspace W_e , and an ordered sequence of harness Sessions. At most one Session is write-capable at a time. A continuation Session receives the same definition, binding, and unpublished workspace; it does not acquire another claim or create an intermediate state.

Process lifetime is therefore not the semantic boundary. A Session may exit without completing the execution, and an execution may continue in another Session. Conversely, a still-live process cannot mutate a state after accepted completion. The Step execution, rather than its operating-system process, owns the single publication decision.

Validated atomic publication

An agent requests completion by declaring outputs, collections, and, for a merge, selected parent identifiers. The engine serializes completion against continuation, suspends writes, and checks four conditions:

1. every required output exists at an allowed path;
2. every required collection is present, including an explicitly empty collection where emptiness is meaningful;
3. selected parents are duplicate-free members of the candidate set frozen in e ; and
4. selected parents collectively cover every exact bound input that requires cross-state merge.

After validation, the engine checkpoints the complete workspace, assigns occurrence identities only to declared publications, freezes collections and fulfillment relationships, constructs the state manifest, and atomically makes the state and successful execution visible. A rejected completion publishes nothing and leaves the execution continuable. Explicit failure closes the execution without a successor and records failure outside the successful state DAG.

The publication rules give the central visibility invariant:

$$\begin{aligned} \text{accepted}(e) &\implies |\text{Successors}(e)| = 1, \\ \neg \text{accepted}(e) &\implies |\text{Successors}(e)| = 0. \end{aligned} \quad (4)$$

No downstream evaluation observes a partially checkpointed state or a successful provenance record without its corresponding state.

Algorithm 1: Reactive grounding, claiming, and publication

Require: Playbook revision P^r , sealed history H

```

1: evaluate once when  $P^r$  is attached
2: loop
3:    $I \leftarrow \text{Snapshot}(H)$ 
4:   for all  $d^r \in P^r$  do
5:     for all  $g \in \text{Ground}(d^r, I)$  do
6:        $r \leftarrow \text{ResolveExactInputs}(d^r, g, I)$ 
7:       if  $r = (b, A, K, o)$  then
8:          $(\text{ready}, u) \leftarrow \text{Ready}(d^r, g, b, I)$ 
9:         if  $\text{ready}$  then
10:           $e \leftarrow \text{ClaimAndCreate}(P^r, d^r, r)$ 
11:          if  $e \neq \text{none}$  then
12:            enqueue  $e$ 
13:          end if
14:        end if
15:      end if
16:    end for
17:  end for
18:  dispatch queued executions within capacity
19:  if no unclaimed binding is eligible and no work is
    queued or running then
20:    report quiescence; continue awaiting events
21:  end if
22:   $x \leftarrow \text{AwaitEvent}()$ 
23:  if  $x = \text{Completion}(e)$  then
24:    if  $\text{ValidateAndPublish}(e, x)$  then
25:      add the sealed successor to  $H$ 
26:    else
27:      keep  $e$  continuable with validation errors
28:    end if
29:  end if
30: end loop

```

Reactive evaluation

Algorithm 1 gives the event-driven reference behavior. Grounding and readiness inspect a consistent snapshot of sealed history. Exact input resolution either waits or returns (b, A, K, o) ; the atomic claim-and-create operation freezes those values in the execution record. Claiming occurs before capacity-limited dispatch, so two evaluator passes cannot create duplicate automatic executions for the same identity. A scheduler chooses when queued work starts but cannot change what the binding means.

The loop in Algorithm 1 represents persistent attachment, not busy polling. Implementations reevaluate on state-sealed and explicit manual events, and otherwise block. A state publication may ground many executions; capacity limits dispatch but does not defer claim identity to a mutable future snapshot. Quiescence is reported after a complete evaluation pass finds no eligible unclaimed grounding and no queued or running execution, regardless of whether the preceding event published a state.

5 The Example as a Concrete Trace

Linear publication boundaries

Let the seed S_0 contain one initial research handoff. Frame executes in an isolated workspace and publishes S_1 with a `research-questions` occurrence. Discover binds that exact occurrence, freezes its source decisions, and publishes S_2 . Until each completion is accepted, the next Step sees neither a half-written question list nor a temporary source map.

The linear trace demonstrates why a full state, rather than an artifact bundle alone, is the unit of publication. An agent may change code, supporting files, and several handoffs together. A complete checkpoint makes the transition auditable and preserves the exact material context for later inspection or a replay attempt, while the artifact map supplies stable logical names.

Fan-out, siblings, and coverage

Here, r_{iX} denotes the appraisal request pairing question i with source X , and S_{iX} is the sibling state publishing the corresponding appraisal. Suppose Discover freezes appraisal requests $r_{1A}, r_{1B}, r_{2B}, r_{2C}$. Four Appraise executions share parent S_2 but receive distinct member bindings. Accepted results create siblings $S_{1A}, S_{1B}, S_{2B}, S_{2C}$ even if their wall-clock completion order differs. The runtime never treats a later-finishing sibling as a descendant of an earlier-finishing sibling.

Synthesize for question 1 binds only successful results fulfilling $\{r_{1A}, r_{1B}\}$. The execution's authorized candidates are frozen when the binding is resolved. At completion, the agent may select the states that actually supply those results; the engine accepts only a duplicate-free set whose union covers both requests. Results for question 2 can serve as a global readiness witness but cannot silently become parents of question 1. The same rule lets Design merge only the exact question synthesizes it uses.

Separating readiness, binding, and lineage prevents three plausible errors. First, a state that merely existed when a Session started does not become a parent. Second, a state used to establish global readiness does not become an input. Third, repeating one parent identifier does not satisfy a two-result merge. Parentage records material dependence rather than temporal proximity.

Another pass

After Design publishes S_D , a follow-up member h_2 names the exact research question that could materially change the design. The member is a new binding, so Frame can execute again under the same definition revision. Subsequent Discover and Appraise executions create new descendants with new occurrence and collection identities, even if a question or source key reuses the same human-readable text.

If Design instead freezes `followup-research` as the empty set, no Frame execution is grounded. The Playbook becomes quiescent after all other claimed work closes. A future manual-publication operation could add another handoff and wake the same Playbook without rewinding or deleting earlier history; its authority and interface remain open.

Semantic obligations exposed by the trace

The example yields five obligations for a compatible runtime:

1. **Atomic visibility:** each accepted execution publishes one full state; rejected or failed work publishes none.
2. **Topology independence:** scheduler order changes latency, not sibling or parent relationships.
3. **Automatic-claim uniqueness:** each automatic claim identity κ is acquired at most once.
4. **Coverage-preserving merge:** selected parents are authorized and collectively supply the bound result occurrences.
5. **Forward recurrence:** repeating a Step definition creates a new descendant; no concrete history edge points backward.

Any runtime claiming compatibility with the model must preserve these obligations. The obligations concern execution provenance and publication topology; the semantic quality of an appraisal or design remains outside the execution specification.

6 Limitations and Open Decisions

This paper specifies execution semantics and illustrates their consequences with a constructed trace. The example demonstrates that the model represents this research-to-design pattern; it does not establish a general expressiveness result, scalability, liveness under failure, agent-output quality, implementation correctness, or operational security. Claims about those properties require separate implementation and evaluation work.

Several mechanisms remain policy rather than semantics. A production specification must settle claim scope across sibling ancestry, retention and retry after failure, deterministic handling of multiple valid fulfillments, empty complete-set behavior, merge workspace conflicts, scheduler fairness, and the representation of manual state creation. Durable publication also requires a concrete transaction and recovery protocol that prevents a checkpoint, manifest, claim, and database record from disagreeing after a crash. External side effects cannot be rolled back merely because filesystem publication fails.

The model currently assumes one authority creates automatic claims for a task. Multiple evaluators would require fencing or a shared transactional store. An isolated workspace is a Playbook visibility boundary, not a security sandbox; network, subprocess, and external-tool confinement remain outside the present semantics. Human interaction is allowed inside a Session and through explicit manual continuations, yet the paper does not define approval policies or how a human selects a historical state in the interface.

Finally, the schematic notation in Appendix A is not a frozen serialization format. The semantic distinctions are the intended contract: finite explicit collections, exact bindings, atomic claims, isolated executions, validated completion, and coverage-preserving lineage. Locking a portable syntax before the recovery and authority decisions would

turn unresolved policy into accidental compatibility requirements.

7 Conclusion

Reactive Playbooks model long-running agent work as provenance-bound transitions over immutable material states. One research-to-design definition consequently expresses linear handoffs, runtime fan-out, concurrent siblings, coverage-preserving merge, and later refinement without a control cursor or cyclic history. The reference algorithm makes the governing obligations explicit: exact grounding, atomic claiming, isolated execution, validated single-state publication, and fail-closed parent coverage. Recovery, authority, resource, retry, and serialization policies remain open, but implementations can resolve those policies against the stable semantic boundary defined here.

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A Schematic Playbook Definition

Listing 1 records a compact definition of the running example. The notation names the semantic relationships used in the paper without freezing a portable serialization format. Agent instructions, harness selection, resource policy, and output paths remain part of a complete Step definition even where the schematic elides them.

Listing 1: Schematic research-to-design Playbook. An initial agenda is represented as the first member of `followup-research`; later Design executions may emit another member or an explicitly empty collection.

```
playbook: research-to-design
collections:
  appraisal-requests: key(question, source)
  appraisal-results: fulfills(appraisal-requests)
  question-synthesis-requests: key(question)
  question-syntheses: fulfills(question-synthesis-requests)
  followup-research: key(handoff)
  framed-questions: fulfills(followup-research)
  design-requests: key(design)

steps:
  frame:
    input: each(followup-research)
    output: framed-questions
  discover:
    input: each(framed-questions)
    emit: [appraisal-requests,
           question-synthesis-requests,
           design-requests]
  appraise:
    input: each(appraisal-requests)
    output: appraisal-results
  synthesize:
    input: each(question-synthesis-requests)
    complete-set: required-appraisals(input)
    when: all-appraisals-from-expansion-fulfilled
    output: question-syntheses
  design:
    input: each(design-requests)
    complete-set: required-syntheses(input)
    output: [design.md, followup-research]
```

B Engine Design Decisions

This appendix records the engine decisions and associated authoring conventions.

1. **No planning search:** the engine rolls out a fixed authored policy rather than searching for plans or inventing collaboration structures.
2. **Fixed process, variable work:** the same Playbook can produce different numbers of executions, branches, and loop passes.
3. **Intelligent transitions:** agents interpret evidence and choose outcomes, allowing the engine's structural rules to stay small.
4. **No general condition language:** domain decisions live in agent instructions and artifacts rather than engine-level expressions.
5. **Artifact-driven dependencies:** declared inputs and outputs determine readiness, independently of document order.
6. **AND-only inputs:** every declared input is required; listing alternatives does not create an OR condition.
7. **At least one output:** successful completion requires one valid assigned artifact, while any additional declared outputs are optional.
8. **No engine-enforced XOR:** publishing two valid alternative outcomes is allowed; instructions must establish any required exclusivity.
9. **Structural validation:** the engine checks publication validity, ownership, and permission without deciding whether the substantive work is good.
10. **Atomic acceptance:** one invalid present assigned output rejects the whole completion attempt rather than accepting a partial subset.
11. **Immutable receipts:** acceptance freezes the publication set, so repeated completion returns the same receipt and late files cannot extend it.
12. **Ordinary mutable files:** accepted artifacts are not protected by immutable byte snapshots or an operating-system write sandbox.
13. **Occurrence identity:** two publications count as distinct events even when their logical names and contents match.
14. **Shared exact roles:** several steps may publish the same exact role, with each eligible occurrence enabling its own continuation.
15. **Unique wildcard ownership:** overlapping wildcard output declarations are rejected to keep collection attribution unambiguous.
16. **Causal input pairing:** companion artifacts must belong together through recorded provenance, not timestamps, filename order, or "latest wins."
17. **Explicit ambiguity errors:** the engine rejects uncertain pairings rather than guessing, zipping alternatives, or trying every combination.
18. **Binding-based deduplication:** observing the same ordinary input binding again does not schedule duplicate work.
19. **Three input modes:** `single` binds one occurrence, each creates per-member work, and `complete` consumes a completed nonempty collection.
20. **One collection input per step:** combining multiple collections requires explicit intermediate steps rather than implicit cross-product joins.
21. **Batch publication:** collections become usable after accepted publication and producer shutdown, rather than streaming files into downstream work.
22. **Accountable completeness:** queued, paused, failed, and finishing workers remain obligations instead of silently disappearing from a merge.
23. **No implicit branch pruning:** an omitted intermediate output does not automatically discharge downstream work that was expected.
24. **No empty consumers:** empty collections create neither item workers nor an empty `complete` execution.
25. **Explicit no-result routing:** a disposition inside a wildcard becomes a member; one outside it can stop or route elsewhere.
26. **Artifact-triggered recurrence:** another pass requires a fresh accepted trigger, not an overwritten file or incremented filename.
27. **No predetermined iteration limit:** loops may continue through fresh publications, with stopping determined by the authored process.
28. **Explicit freshness structure:** authors must put changing work inside the repeated dependency chain; parser acceptance alone does not prove freshness.
29. **Explicit accumulated context:** agents carry useful earlier findings into new handoffs; the engine does not automatically summarize task history.
30. **No automatic success inference:** having no eligible work does not prove that the human's objective was achieved.
31. **Execution-scoped completion permission:** a human grant authorizes that execution and owner to finish, not particular immutable artifact contents.
32. **Completion differs from shutdown:** downstream work waits for confirmed producer exit and output drain after acceptance.
33. **One shared task worktree:** engine-managed coding is serialized instead of giving every execution an isolated workspace and later merging changes.
34. **Exact coding inputs:** collection results must be consolidated before coding, although coding steps may publish collections for subsequent work.
35. **Task-level concurrency:** the default cap is ten live sessions; excess work queues, and waiting or finishing sessions retain capacity.
36. **No automatic retries or takeover:** recovery requires explicit ownership changes and proof that the previous owner stopped.
37. **Manual work stays separate:** manual publications do not automatically satisfy ordinary graph dependencies or missing worker obligations.

38. **Pinned task definitions:** library edits affect future selections rather than silently changing an existing task's Playbook.
39. **Daemon-owned execution truth:** durable execution records drive scheduling; session metadata and UI state are repairable projections.
40. **File-backed persistence:** the engine uses durable local records rather than adding a database or distributed execution coordinator.
41. **Persist intent before effects:** executions and output paths are reserved before launch, while partial creation failures remain inspectable.
42. **Names are presentation:** depth numbers and collision suffixes help people navigate artifacts but never determine provenance or scheduling.
43. **Strict definitions:** unknown fields and unsupported syntax fail explicitly rather than acquiring guessed meanings or compatibility fallbacks.
44. **Narrow harness support:** model selection is configurable, but current graph execution supports OMP rather than arbitrary harness integrations.